

Surveillance Using Raspberry Pi Camera

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This surveillance system detects motion of any object within its camera's range, captures the image and stores it in Raspberry Pi. You can retrieve images from Rasp-

berry Pi any time.

Since Raspberry Pi camera is static-sensitive, take good grounding measures before you touch it. Connect the camera as shown in Fig. 1.

Installation of camera

Open Raspi config utility using the command:

```
$ sudo raspi-config
```

Now, enable the camera as shown in Fig. 2.

If your firmware is old, you need to update and upgrade it using the command mentioned below to see the camera enable option:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

After enabling and connecting the camera to Raspberry Pi, use following commands to check if it is working well. The first command is raspistill, which is used to take still pictures. For example, if you want to capture an image, type the following command (also see Fig. 3):

```
$ raspistill -v -o image.jpg
```

If red LED of the Raspberry Pi camera glows for some time, it shows that the image has been recorded. You may check the image under Home folder in Raspberry Pi to ensure that the camera is properly installed.

Capturing motion

We have two methods for motion detection. First method uses the light-weight motion detection Python script while

the second method uses 'motion' software to detect motion and capture images.



Method 1: Using Python script.

Python script depends on Python Imaging Library, which is meant for analysing and manipulating images. So we have to install the library using the following command (also see Fig. 4):

```
$ sudo apt-get install python-imaging-tk
```

We can get the script from the link mentioned below:

<https://github.com/skl/raspberry-pi-cam/archive/master.zip>

Make the script executable by using the command (also see Fig. 5):

```
$ chmod +x picam.py
```

The script is designed to store images in a directory named picam under Home directory, so create it as shown below before executing the script:

```
$ mkdir ~/picam
```

We shall be ready to run the script after giving the following command:

```
$ ./picam.py
```

The script will turn on red LED of the Raspberry Pi camera and start taking low-resolution images. It will then compare them and look for movement by comparing the pixels in the images. If it detects any changes in the pixels, the script will capture a higher-resolution image.

The script is written in such a way that it stores only high-resolution images. All these images are saved in the ~/picam folder which we had created. Note that, if the camera is placed in a windy area then set the threshold variable in config file to a higher value than the default.

For running the script during booting, we need an init script that runs the picam.py script and kills it before shutting down the Raspberry Pi. To get the script, issue the command given below:

```
$ wget http://pastebin.com/AfqbjQrb
-o picam_init
```

Move the script into its correct



Fig. 1: Raspberry Pi and the camera



Fig. 2: Enabling the camera



Fig. 3: Capturing an image



Fig. 4: Installing Python script

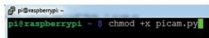


Fig. 5: Making the script executable